

TRUNG TIN BUI

AI Engineer · LLM Systems · Research Software

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EDUCATION

International University, Vietnam National University Ho Chi Minh City

Ho Chi Minh City, Vietnam

Bachelor of Engineering in Biomedical Engineering

2020 - Jan 2025

- GPA: 3.54/4.00; thesis score: 94/100; TOEIC: 870.
- Student Best Poster Presentation, 10th International Conference on the Development of Biomedical Engineering in Vietnam, 2024.

EXPERIENCE

LOGINEX Technology

Remote

AI Engineer

2026 - Present

- Engineer a license-plate OCR pipeline that converts vehicle imagery into structured plate data for operational workflows.
- Build an enterprise RAG system with role-based and attribute-based access control (RBAC/ABAC), ensuring retrieval respects user roles, resource attributes, and business context.

viAct

Vietnam

Gen AI Engineer

Mar 2025 - Jul 2026

- Built production FastAPI, Celery, and React services for construction operations, connecting CCTV feeds, documents, and workflow tools through authenticated APIs and containerized deployments.
- Engineered Slack and Jira agents with tool actions, long-term memory, and human-in-the-loop approval, reducing task creation and tracking time by approximately 70%.
- Designed an end-to-end CCTV retrieval system that ingested and chunked video, generated embeddings, indexed footage in ChromaDB, and resolved natural-language queries to timestamped segments with camera and time-range filters.
- Deployed enterprise RAG and document-intelligence pipelines with Elasticsearch and PostgreSQL; integrated custom handwriting OCR that reached approximately 90% accuracy on complex construction forms.
- Integrated WhisperX for audio transcription and alignment, reducing transcription failures by approximately 40% in operational media workflows.
- Created a full SAM workflow with a custom annotation interface, Google Drive synchronization, automatic SAM-to-YOLO dataset conversion, and model-serving scripts for construction safety use cases.
- Operated live RTSP vision workloads on Ray Serve and introduced automatic unloading for inactive models to release constrained GPU capacity.
- Quantized Qwen-VL models, converted YOLO and OCR models to ONNX, and integrated vLLM, reducing GPU memory use by approximately 60% and increasing inference throughput by approximately 3x.

GrantForward · Cazoodle

Remote

Research Software Engineer

Sep 2025 - Oct 2025

- Architected a research discovery prototype that indexed large academic corpora and combined semantic vector retrieval with RAG for complex scientific questions.
- Compared semantic results with keyword baselines and documented retrieval-quality and maintainability trade-offs for a distributed product team.

FPT Digital

Ho Chi Minh City, Vietnam

AI Engineer Intern, Full-time

Jan 2025 - Mar 2025

- Translated stakeholder requirements into an internal RAG assistant for staff knowledge retrieval and iterated on answer quality from operational feedback.
- Researched and prototyped enterprise AI workflows while collaborating with engineering and business stakeholders.

IVS Joint Stock Company
AI Engineer Intern, Hybrid

Ho Chi Minh City, Vietnam
Jul 2023 - Aug 2024

- Collected, cleaned, and preprocessed large Vietnamese medical text datasets for model training and evaluation.
- Fine-tuned Vistral, PhoGPT, Llama 2, and SeaLLM with LoRA/QLoRA to develop a specialized medical assistant.
- Trained and evaluated a BERT classifier that achieved 83% accuracy across 23 medical classes.

Brain AI Lab 513, School of Biomedical Engineering, VNU-HCMC
Research Assistant

Ho Chi Minh City, Vietnam
Aug 2021 - Feb 2025

- Applied U-Net, ResNet-50, VGG-16/19, CNN, and YOLOv5 to medical image segmentation and classification, reaching 97% accuracy on skin-disease classification.
- Preprocessed and analyzed brain MRI images for CNN-based Alzheimer's disease classification.
- Built a LangChain and Elasticsearch RAG system for mental-health information retrieval.

Teaching Assistant, AI for Healthcare

Aug 2023 - Aug 2024

- Mentored more than 50 biomedical engineering students and prepared coding laboratories, technical documents, quizzes, and assignments on applied machine learning.

SELECTED PROJECTS

Efficient Medical LLM Fine-Tuning — Optimized LLaMa2-7B medical QA with QLoRA; achieved BERTScore F1 0.841 versus Gemini 0.849 and GPT 0.855 while retaining a smaller model and faster fine-tuning.

Gas Store Operations ERP — Shipped a FastAPI/PostgreSQL API, React web app, and offline-first Expo Android client with dual-mode JWT auth, backend RBAC, idempotent SQLite outbox sync, order-level debt, inventory, cylinder audits, Docker Compose, and CI.

Mental Health RAG Platform — Created a full-stack retrieval experience with React, LangChain, FastAPI, and Elasticsearch to answer mental-health questions from indexed knowledge.

HI-CARD Biomedical Capstone — Developed a healthcare capstone with a Python backend and companion information interface.

VHCor Vietnamese Healthcare Corpus — Co-developed a dataset for Vietnamese medical department recognition and published its methodology and benchmark at an international conference.

PUBLICATIONS

- “Optimizing LLaMa2 Fine-Tuning with LoRA and Quantization for Medical Question Answering,” BME 2024, IFMBE Proceedings, Springer, online 2025.
- “VHCor: Vietnamese HealthCare Corpus - A Comprehensive Dataset for Vietnamese Clinic Recognition,” International Conference on Health Science and Technology, 2023.

TECHNICAL SKILLS

Applied AI: RAG architecture, AI agents, LLM fine-tuning, LoRA/QLoRA, retrieval evaluation, prompt engineering, OCR, WhisperX, document intelligence.

Computer Vision: YOLO, SAM, Qwen-VL, OpenCV, U-Net, ResNet, RTSP video pipelines.

Software and Data: Python, FastAPI, Django, React, TypeScript, PostgreSQL, MongoDB, Elasticsearch, ChromaDB, Celery, REST APIs.

Infrastructure: PyTorch, Hugging Face, Ray Serve, vLLM, ONNX, Docker, Git.

CERTIFICATIONS AND AWARDS

Certifications: Claude Code in Action; AWS Cloud Technical Essentials; Neural Networks and Deep Learning; TOEIC Speaking and Writing 310.

Awards: Student Best Poster Presentation, BME10 (2024); The Most Favorite Idea Award.